

Introduction to Machine Learning

Machine Learning is a branch of Artificial Intelligence that enables computers to learn patterns from data without being explicitly programmed. Machine Learning algorithms improve their performance as they are exposed to more data.

Types of Machine Learning

Supervised Learning

Supervised learning uses labeled datasets to train predictive models.

Common algorithms include:

- Linear Regression
- Logistic Regression
- Decision Trees
- Random Forest
- Support Vector Machine (SVM)

Applications include spam detection, house price prediction, and medical diagnosis.

Unsupervised Learning

Unsupervised learning discovers hidden patterns in unlabeled data.

Popular algorithms include:

- K-Means Clustering
- DBSCAN
- Hierarchical Clustering
- Principal Component Analysis (PCA)

Applications include customer segmentation and anomaly detection.

Reinforcement Learning

Reinforcement Learning trains an agent to make decisions by interacting with an environment. The agent receives rewards or penalties based on its actions.

Applications include:

- Robotics
- Self-driving cars
- Game playing
- Resource optimization

Data Preprocessing

Data preprocessing is one of the most important steps in Machine Learning.

Common preprocessing techniques include:

- Handling missing values
- Removing duplicate records
- Encoding categorical variables
- Feature scaling
- Outlier detection
- Feature engineering

Popular scaling methods include:

- StandardScaler
- MinMaxScaler

Outlier detection can be performed using:

- IQR Method
- Z-Score

Model Evaluation

Classification models are evaluated using metrics such as:

- Accuracy
- Precision
- Recall
- F1 Score

Regression models are evaluated using:

- Mean Absolute Error (MAE)
- Mean Squared Error (MSE)
- Root Mean Squared Error (RMSE)

Cross-validation is commonly used to estimate model performance.

Deployment

Machine Learning models can be deployed using:

- FastAPI
- Flask
- Docker
- Kubernetes

Cloud deployment platforms include:

- Google Cloud Platform
- Microsoft Azure
- Amazon Web Services (AWS)